TEST REPORT

Application No.:	KSEM2204000432AT
Applicant:	Richbeam (Suzhou) Technology Co.,LTD
Address of Applicant:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Manufacturer:	Richbeam (Suzhou) Technology Co.,LTD
Address of Manufacturer:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Factory:	Richbeam (Suzhou) Technology Co.,LTD
Address of Factory:	NO.788, Southeast Avenue, Changshu high tech Industrial Development zone, Jiangsu, China
Equipment Under Test (EUT):	
EUT Name:	2D TOF LIDAR
Model No.:	LakiBeam 1, LakiBeam 1L ♣
♣	Please refer to section 2 of this report which indicates which model was actually tested and which were electrically identical.
Trade Mark:	RICHBEAM
Standard(s) :	EN IEC 61000-6-3: 2021 EN IEC 61000-6-1: 2019
Date of Receipt:	2022-04-12
Date of Test:	2022-04-25 to 2022-04-28
Date of Issue:	2022-05-05
Test Result:	**Pass***

* In the configuration tested, the EUT complied with the standards specified above.

Eric Lin
EMC Laboratory Manager

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Revision Record				
Version	**Chapter**	**Date**	**Modifier**	**Remark**
01		2022-05-05		Original

Authorized for issue by:				
		Lee Li **Lee Li/Project Engineer**		
		Eric Lin **Eric Lin/Reviewer**		

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2 Test Summary

Emission Part				
Item	**Standard**	**Method**	**Requirement**	**Result**
Conducted Emissions at AC Mains Power Port (150kHz-30MHz)	EN IEC 61000-6-3: 2021	CISPR 16-2-1	Table 4.3	Pass
Conducted Emissions at Wired Network Port (150kHz-30MHz)		CISPR 32	Table 6	Pass
Radiated Emissions (30MHz-1GHz)		CISPR 16-2-3	Table 3.1	Pass
Harmonic Current Emission		EN IEC 61000-3-2: 2019+A1:2021	Class A	Pass
Voltage Fluctuations and Flicker		EN 61000-3-3:2013+A1:2019	Clause 5	Pass

Immunity Part				
Item	**Standard**	**Method**	**Requirement**	**Result**
Conducted Immunity at AC Mains Power Port (150kHz-80MHz)	EN IEC 61000-6-1: 2019	EN 61000-4-6:2014	3Vrms (emf),80%,1kHz Amp. Mod.	Pass
Conducted Immunity at Signal Port (150kHz-80MHz)		EN 61000-4-6:2014	3Vrms (emf),80%,1kHz Amp. Mod.	Pass
Electrical Fast Transients Burst at AC Mains Power Port		EN 61000-4-4:2012	1kV 5/50ns Tr/Td 5kHz/100kHz Repetition Frequency	Pass
Electrical Fast Transients Burst at Signal Port		EN 61000-4-4:2012	1kV 5/50ns Tr/Td 5kHz/100kHz Repetition Frequency	Pass
Electrostatic Discharge		EN 61000-4-2:2009	±4kV Contact Discharge, ±8kV Air Discharge	Pass
Radiated Immunity (80MHz-6GHz)		EN IEC 61000-4-3: 2020	3V/m, 80%, 1kHz Amp. Mod.	Pass
Surge at AC Mains Power Port		EN 61000-4-5:2014+A1:2017	1.2/50μs Tr/Td 1kV Line to Line	Pass
Surge at Signal Port		EN 61000-4-5:2014+A1:2017	1.2/50μs Tr/Td 1kV Line to Ground	Pass
Voltage Dips and Interruptions		EN IEC 61000-4-11:2020	0 % UT for 0.5cycle 0 % UT for 1cycle 70 % UT for 25cycles 0 % UT for 250cycles UT is Supply Voltage	Pass

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Immunity Part				
Item	**Standard**	**Method**	**Requirement**	**Result**
Power Frequency Magnetic Field		EN 61000-4-8:2010	50Hz 3A/m	Pass

Declaration of EUT Family Grouping:

There are series models mentioned in this report, and they are the Identical in electrical and electronic characters.Only the model LakiBeam 1L was tested since their differences were the model number, trade name,Color and appearance.

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4 General Information

4.1 Details of E.U.T.

Power supply:	DC 9~36V
Test Voltage:	AC 230V/50Hz

4.2 Description of Support Units

Description	Manufacturer	Model No.	Serial No.
AC Adapter	DVE	DSA-12G-12FEU	/
Notebook	XIAOMI	Pro15	00342-35744-35614-AAOEM

4.3 Measurement Uncertainty

No.	Item	Measurement Uncertainty
1	Conducted Emission at mains port using AMN	2.4dB (9kHz to 150kHz)
		2.2dB (150kHz to 30MHz)
2	Conducted Emission at telecommunication port using AAN	4.0 dB (150kHz to 30MHz)
3	Radiated Power	3.2dB
4	Radiated Emission (10m)	4.1 dB
5	Radiated Emission (3m)	4.6 dB (30MHz-1GHz)
		5.0dB (1GHz-6GHz)
		5.2dB (6GHz-18GHz)
		5.3dB (18GHz-40GHz)

Note: The measurement uncertainty represents an expanded uncertainty expressed at approximately the 95% confidence level using a coverage factor of k=2.

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4.4 Test Location

All tests were performed at:

Compliance Certification Services (Kunshan) Inc.

No.10 Weiye Rd, Innovation park, Eco&Tec, Development Zone, Kunshan City, Jiangsu, China.

Tel: +86 512 5735 5888 Fax: +86 512 5737 0818

No tests were sub-contracted.

4.5 Test Facility

The test facility is recognized, certified, or accredited by the following organizations:

• CNAS (No. CNAS L4354)

CNAS has accredited Compliance Certification Services (Kunshan) Inc. to ISO/IEC 17025:2017 General Requirements for the Competence of Testing and Calibration Laboratories (CNAS-CL01 Accreditation Criteria for the Competence of Testing and Calibration Laboratories) for the competence in the field of testing.

• A2LA (Certificate No. 2541.01)

Compliance Certification Services (Kunshan) Inc. is accredited by the American Association for Laboratory Accreditation (A2LA). Certificate No. 2541.01.

• FCC (Designation Number: CN1172)

Compliance Certification Services (Kunshan) Inc. has been recognized as an accredited testing laboratory.

Designation Number: CN1172.

• ISED (CAB identifier: CN0072)

Compliance Certification Services (Kunshan) Inc. has been recognized by Innovation, Science and Economic Development Canada (ISED) as an accredited testing laboratory.

Company Number: 2324E

• VCCI (Member No.: 1938)

The 3m and 10m Semi-anechoic chamber and Shielded Room of Compliance Certification Services (Kunshan) Inc. has been registered in accordance with the Regulations for Voluntary Control Measures with Registration No.: R-20134, R-11600, C-11707, T-11499, G-10216 respectively.

4.6 Deviation from Standards

None

4.7 Abnormalities from Standard Conditions

None

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5 Equipment List

Conducted Emissions at AC Mains Power Port (150kHz-30MHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMI TEST RECEIVER	R&S	ESCI	100781	01/22/2022	01/21/2023
TWO-LINE V-NETWORK	R&S	ENV216	101604	10/12/2021	10/11/2022
V (V-LISN)	SCHWARZBECK	NNLK 8129	8129-143	10/12/2021	10/11/2022
Pulse LIMITER	R&S	ESH3-Z2	100609	01/22/2022	01/21/2023
Software	Faratronic	EZ_EMC-v 3A1	N/A	N/A	N/A

Conducted Emissions at Wired Network Port (150kHz-30MHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMI TEST RECEIVER	R&S	ESCI	100781	01/22/2022	01/21/2023
TWO-LINE V-NETWORK	R&S	ENV216	101604	10/12/2021	10/11/2022
V (V-LISN)	SCHWARZBECK	NNLK 8129	8129-143	10/12/2021	10/11/2022
Pulse LIMITER	R&S	ESH3-Z2	100609	01/22/2022	01/21/2023
CISPR22 FOUR BALANCED TELECOM PARIS ISN	FCC	FCC-TLISN-T2-02	20625	01/22/2022	01/21/2023
COUPLING AND DECOUPLING NETWORK	TESEQ	ISN ST08	31272	01/22/2022	01/21/2023
IMPEDANCE STABILIZATION NETWORK	TESEQ	ISN T800	34450	01/22/2022	01/21/2023
IMPEDANCE STABILIZATION NETWORK	TESEQ	ISN T8-CAT6	38913	01/22/2022	01/21/2023
RF CURRENT PROBE	FCC	F-65A	146	01/22/2022	01/21/2023
Software	Faratronic	EZ_EMC-v 3A1	N/A	N/A	N/A

Radiated Emissions (30MHz-1GHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMI Test Receiver	R&S	ESCI	101378	10/11/2021	10/10/2022
Antenna	TESEQ	CBL 6112D	35403	03/05/2022	03/04/2024
Software	Faratronic	EZ_EMC v 3A1	N/A	N/A	N/A

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Harmonic Current Emission					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Harmonic & Flicker Tester	SCHAFFNER	CCN 1000-1	72585	10/18/2021	10/17/2022
AC Power Source	SCHAFFNER	NSG 1007	54788	10/18/2021	10/17/2022
Software	TESEQ	CTS4-v 4.24.0	N/A	N/A	N/A
Harmonic/Flicker Analyzer	KIKUSUI	KHA3000	AS000065	04/01/2022	03/31/2023
Line Impedance Network	KIKUSUI	SPEC71116	/	04/01/2022	03/31/2023
Switcher	KIKUSUI	SPEC71092	/	08/24/2021	08/23/2022
AC Power Supply(Master)	KIKUSUI	PCR24000WE 2	AR001701	04/01/2022	03/31/2023
AC Power Supply(Slave)	KIKUSUI	PCR24000WE 2	AN002885	04/01/2022	03/31/2023
Software	KIKUSUI	HarmoCapture 3-vv 2.5.2.00	N/A	N/A	N/A

Voltage Fluctuations and Flicker					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Harmonic & Flicker Tester	SCHAFFNER	CCN 1000-1	72585	10/18/2021	10/17/2022
AC Power Source	SCHAFFNER	NSG 1007	54788	10/18/2021	10/17/2022
Software	TESEQ	CTS4-v 4.24.0	N/A	N/A	N/A
Harmonic/Flicker Analyzer	KIKUSUI	KHA3000	AS000065	04/01/2022	03/31/2023
Line Impedance Network	KIKUSUI	SPEC71116	/	04/01/2022	03/31/2023
Switcher	KIKUSUI	SPEC71092	/	08/24/2021	08/23/2022
AC Power Supply(Master)	KIKUSUI	PCR24000WE 2	AR001701	04/01/2022	03/31/2023
AC Power Supply(Slave)	KIKUSUI	PCR24000WE 2	AN002885	04/01/2022	03/31/2023
Software	KIKUSUI	HarmoCapture 3-vv 2.5.2.00	N/A	N/A	N/A

Conducted Immunity at AC Mains Power Port (150kHz-80MHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Test System for Conducted and Radiated Immunity	TESEQ	NSG 4070B	44279	01/22/2022	01/21/2023
Amplifier	TESEQ	SCCXE75	U2060-0916	01/22/2022	01/21/2023
EM-Koppelzange	SCHAFFNER	KEMZ 801	17629	01/22/2022	01/21/2023
Attenuator	EURO MC	7860	2226	04/01/2022	03/31/2023

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		ORGEVAL			
Directional Coupler	HIGH POWER	C21A8	450	10/12/2021	10/11/2022
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M216	16399	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M316	16935	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	TESEQ	CDN S751	31084	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	TESEQ	CDN M116	35358	04/01/2022	03/31/2023
CDN	TESEQ	CDN T2-10S	39812	04/01/2022	03/31/2023
CDN	TESEQ	CDN T4-10S	37195	04/01/2022	03/31/2023
CDN	TESEQ	CDN T8-10S	42002	04/01/2022	03/31/2023
Current Clamp	TESEQ	IP-DR250	1609-1936	04/01/2022	03/31/2023
CDN	TESEQ	CDN M432	55985	01/22/2022	01/21/2023
CDN	TESEQ	CDN M432-3LN	56245	01/22/2022	01/21/2023
CDN	TESEQ	CDN M532	56180	01/26/2022	01/25/2023
CDN	TESEQ	CDN M232	45974	04/01/2022	03/31/2023
CDN	TESEQ	CDN M332	45360	04/01/2022	03/31/2023
Software	TESEQ	NSG 4070-v 1.3.0.1	N/A	N/A	N/A

Conducted Immunity at Signal Port (150kHz-80MHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Test System for Conducted and Radiated Immunity	TESEQ	NSG 4070B	44279	01/22/2022	01/21/2023
Amplifier	TESEQ	SCCXE75	U2060-0916	01/22/2022	01/21/2023
EM-Koppelzange	SCHAFFNER	KEMZ 801	17629	01/22/2022	01/21/2023
Attenuator	EURO MC	7860 ORGEVAL	2226	04/01/2022	03/31/2023
Directional Coupler	HIGH POWER	C21A8	450	10/12/2021	10/11/2022
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M216	16399	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	SCHAFFNER	CDN M316	16935	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	TESEQ	CDN S751	31084	04/01/2022	03/31/2023
CDN（Coupling and Decoupling Network)	TESEQ	CDN M116	35358	04/01/2022	03/31/2023
CDN	TESEQ	CDN T2-10S	39812	04/01/2022	03/31/2023

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CDN	TESEQ	CDN T4-10S	37195	04/01/2022	03/31/2023
CDN	TESEQ	CDN T8-10S	42002	04/01/2022	03/31/2023
Current Clamp	TESEQ	IP-DR250	1609-1936	04/01/2022	03/31/2023
CDN	TESEQ	CDN M432	55985	01/22/2022	01/21/2023
CDN	TESEQ	CDN M432-3LN	56245	01/22/2022	01/21/2023
CDN	TESEQ	CDN M532	56180	01/26/2022	01/25/2023
CDN	TESEQ	CDN M232	45974	04/01/2022	03/31/2023
CDN	TESEQ	CDN M332	45360	04/01/2022	03/31/2023
Software	TESEQ	NSG 4070-v 1.3.0.1	N/A	N/A	N/A

Electrical Fast Transients Burst at AC Mains Power Port					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	Transient2000	TRA2006_F-S-T-D-R-1503	12/19/2021	12/18/2022
Coupling Network	EMC PARTNER	CN-EFT1000	CN-EFT1000_1535	12/19/2021	12/18/2022
Burst Generator	SANKI	EFT-0404S	040720051E	04/01/2022	03/31/2023
Coupling and Decoupling Network	SANKI	CDN-4350	043520001J	04/01/2022	03/31/2023

Electrical Fast Transients Burst at Signal Port					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	Transient2000	TRA2006_F-S-T-D-R-1503	12/19/2021	12/18/2022
Coupling Network	EMC PARTNER	CN-EFT1000	CN-EFT1000_1535	12/19/2021	12/18/2022
Burst Generator	SANKI	EFT-0404S	040720051E	04/01/2022	03/31/2023
Coupling and Decoupling Network	SANKI	CDN-4350	043520001J	04/01/2022	03/31/2023

Electrostatic Discharge					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
ESD Simulator	EM TEST	DITO 509030	V0936105118	01/26/2022	01/25/2023

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Radiated Immunity (80MHz-6GHz)					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Synthesized Signal Generator	AGILENT	83732B	US37101915	01/22/2022	01/21/2023
Laser probe interface	AR Worldwide	FI7000	354057	04/01/2022	03/31/2023
E-Field Sensor	AR Worldwide	FL7006 100K-6G	342390	04/01/2022	03/31/2023
Amplifier Research (80~1000MHz 150w)	AR Worldwide	150W1000M1	320947	10/18/2021	10/17/2022
Amplifier Research (1~6GHz 50w)	AR Worldwide	50S1G6M1	342706	10/18/2021	10/17/2022
Dual Directional Coupler (1-11G)	AR Worldwide	C1-A47NFNF 35dB	1	N.C.R	N.C.R
Dual Directional Coupler (80~1000MHz 400w)	AR Worldwide	DC6180	302211	N.C.R	N.C.R
RF POWER METER	BOONTON	4232A-01	1614	01/22/2022	01/21/2023
POWER SENSOR	BOONTON	51085	33488	01/22/2022	01/21/2023
POWER SENSOR	BOONTON	51085	33489	01/22/2022	01/21/2023
Antenna	AR Worldwide	TP1000A	302624	N.C.R	N.C.R
Software	AR	emc ware-v 3.2.0.4	N/A	N/A	N/A

Surge at AC Mains Power Port					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	F-S-T-D-R-1503	12/19/2021	12/18/2022
Coupling and Decoupling Network	EMC PARTNER	CDN-UTP8	1511	12/19/2021	12/18/2022
Surge Generator	SANKI	LSG-0506S	050920052J	04/01/2022	03/31/2023
Coupling and Decoupling Network	SANKI	CDN-5350	053520001J	06/08/2021	06/07/2022

Surge at Signal Port					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	F-S-T-D-R-1503	12/19/2021	12/18/2022
Coupling and Decoupling Network	EMC PARTNER	CDN-UTP8	1511	12/19/2021	12/18/2022
Surge Generator	SANKI	LSG-0506S	050920052J	04/01/2022	03/31/2023

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Coupling and Decoupling Network	SANKI	CDN-5350	053520001J	06/08/2021	06/07/2022

Voltage Dips and Interruptions					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	TRA2006_F-S-T-D-R-1503	12/19/2021	12/18/2022
Switcher	KIKUSUI	SPEC71092	/	08/24/2021	08/23/2022
AC Power Supply(Master)	KIKUSUI	PCR24000WE2	AR001701	04/01/2022	03/31/2023
AC Power Supply(Slave)	KIKUSUI	PCR24000WE2	AN002885	04/01/2022	03/31/2023
Software	KIKUSUI	Quick Immunity Sequencer 2-v 4.0.3.02	N/A	N/A	N/A

Power Frequency Magnetic Field					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
EMC Immunity Tester	EMC PARTNER	TRA2006	TRA2006_F-S-T-D-R-1503	12/19/2021	12/18/2022
Inductive Standard Coil	EMC PARTNER	MF1000-1	MF1000-1-1514	12/19/2021	12/18/2022

General used equipment					
Equipment	**Manufacturer**	**Model No**	**Inventory No**	**Cal Date**	**Cal Due Date**
Digital Pressure Meter	Mengde	DYM3	CZ750023	02/01/2022	01/31/2023
Temperature & Humidity Recorder	Anymetre	TH603	CZ720001-1 CZ720001-2 CZ720001-3 CZ720001-4 CZ720001-5 CZ720001-6 CZ720001-7	10/14/2021	10/13/2022

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6 Emission Test Results

6.1 Conducted Emissions at AC Mains Power Port (150kHz-30MHz)

Test Requirement: EN IEC 61000-6-3: 2021

Test Method: CISPR 16-2-1

Limit:

0.15M-0.5MHz	66dB(μV)-56dB(μV) quasi-peak, 56dB(μV)-46dB(μV) average
0.5M-5MHz	56dB(μV) quasi-peak, 46dB(μV) average
5M-30MHz	60dB(μV) quasi-peak, 50dB(μV) average
Detector:	Peak for pre-scan (9kHz resolution bandwidth) 0.15M to 30MHz

6.1.1 E.U.T. Operation

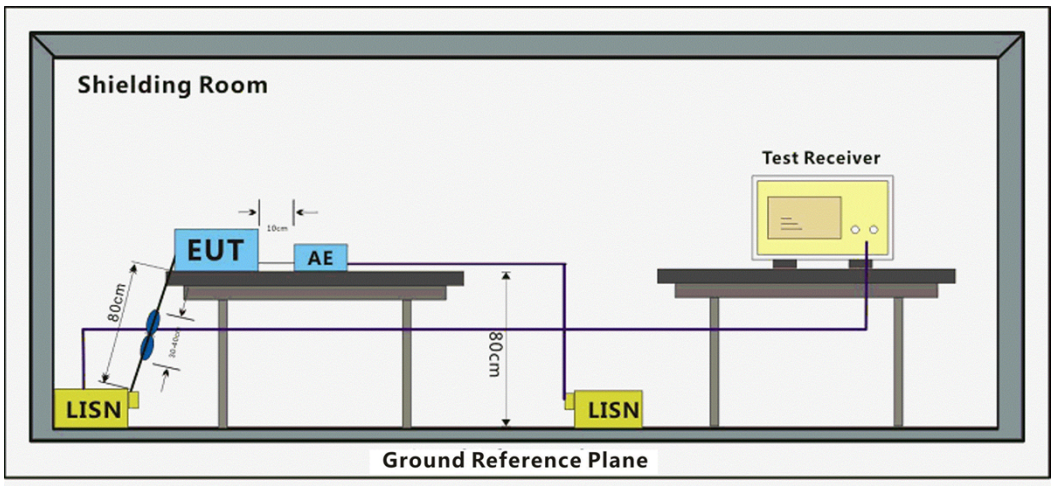
Operating Environment:

Temperature: 26.6 °C Humidity: 55.9 % RH Atmospheric Pressure: 1010 mbar

6.1.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

6.1.3 Test Setup Diagram



6.1.4 Measurement Procedure and Data

Frequency range: 150KHz-30MHz

An initial pre-scan was performed with peak detector.Quasi-Peak or Average measurement were performed at the frequencies with maximized peak emission were detected.

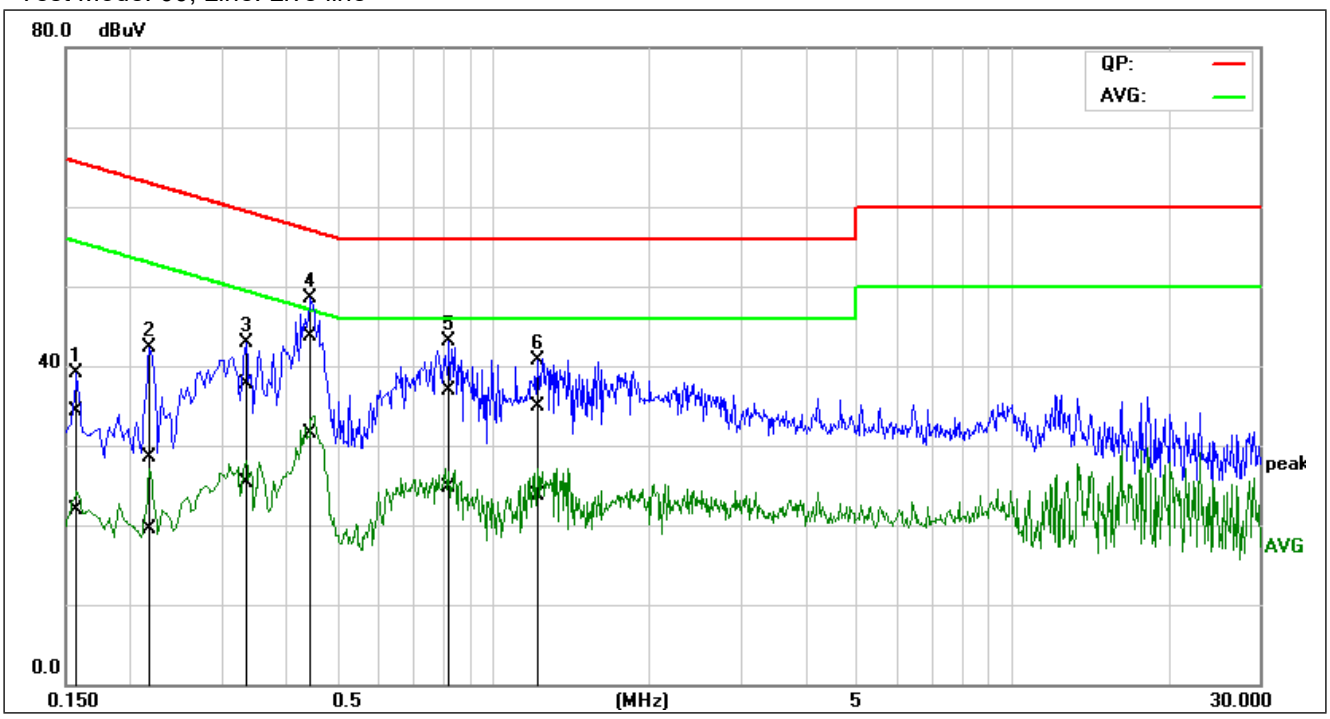
The red line show in graphic is the limit in standard used in this section.

Measured Level = Read level + Cable Loss + LISN Factor

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Test Mode: 00; Line: Live line

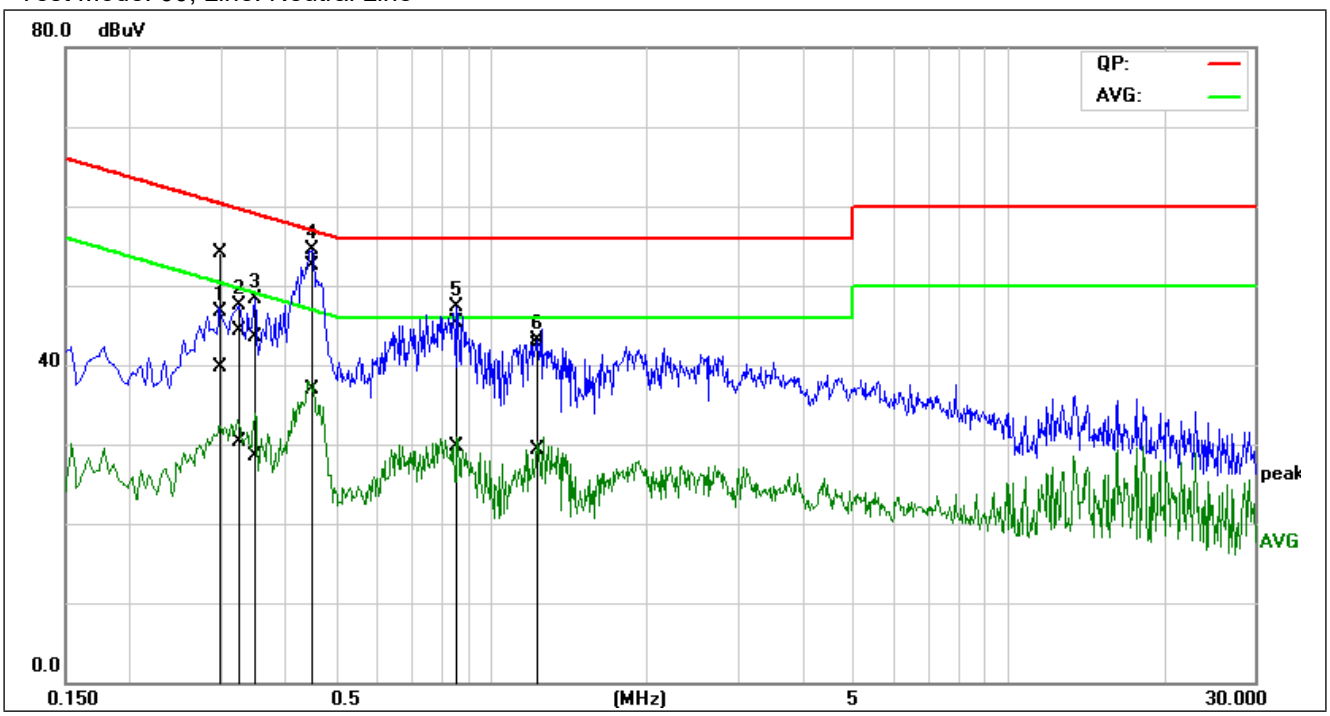


No.	Frequency	QuasiPeak reading	Average reading	Correction factor	QuasiPeak result	Average result	QuasiPeak limit	Average limit	QuasiPeak margin	Average margin	Remark
	(MHz)	(dBuV)	(dBuV)	(dB)	(dBuV)	(dBuV)	(dBuV)	(dBuV)	(dB)	(dB)	
1	0.1582	14.79	2.40	19.46	34.25	21.86	65.56	55.56	-31.31	-33.70	Pass
2	0.2167	9.06	0.14	19.46	28.52	19.60	62.94	52.94	-34.42	-33.34	Pass
3	0.3326	18.16	5.85	19.47	37.63	25.32	59.39	49.39	-21.76	-24.07	Pass
4*	0.4455	24.26	12.10	19.47	43.73	31.57	56.96	46.96	-13.23	-15.39	Pass
5	0.8248	17.48	5.32	19.48	36.96	24.80	56.00	46.00	-19.04	-21.20	Pass
6	1.2229	15.47	4.27	19.48	34.95	23.75	56.00	46.00	-21.05	-22.25	Pass

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Test Mode: 00; Line: Neutral Line



No.	Frequency	QuasiPeak reading	Average reading	Correction factor	QuasiPeak result	Average result	QuasiPeak limit	Average limit	QuasiPeak margin	Average margin	Remark
	(MHz)	(dBuV)	(dBuV)	(dB)	(dBuV)	(dBuV)	(dBuV)	(dBuV)	(dB)	(dB)	
1	0.3013	34.53	20.17	19.50	54.03	39.67	60.21	50.21	-6.18	-10.54	Pass
2	0.3233	24.70	10.72	19.51	44.21	30.23	59.62	49.62	-15.41	-19.39	Pass
3	0.3509	23.98	9.01	19.51	43.49	28.52	58.94	48.94	-15.45	-20.42	Pass
4*	0.4521	32.96	17.36	19.55	52.51	36.91	56.84	46.84	-4.33	-9.93	Pass
5	0.8546	25.72	10.08	19.59	45.31	29.67	56.00	46.00	-10.69	-16.33	Pass
6	1.2240	22.82	9.76	19.61	42.43	29.37	56.00	46.00	-13.57	-16.63	Pass

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6.2 Conducted Emissions at Wired Network Port (150kHz-30MHz)

Test Requirement: EN IEC 61000-6-3: 2021

Test Method: CISPR 32

Limit:

0.15M-0.5MHz	84dB(μV)-74dB(μV) quasi-peak, 74dB(μV)-64dB(μV) average
0.5M-30MHz	74dB(μV) quasi-peak, 64dB(μV) average
Detector:	Peak for pre-scan (9kHz resolution bandwidth) 0.15M to 30MHz

6.2.1 E.U.T. Operation

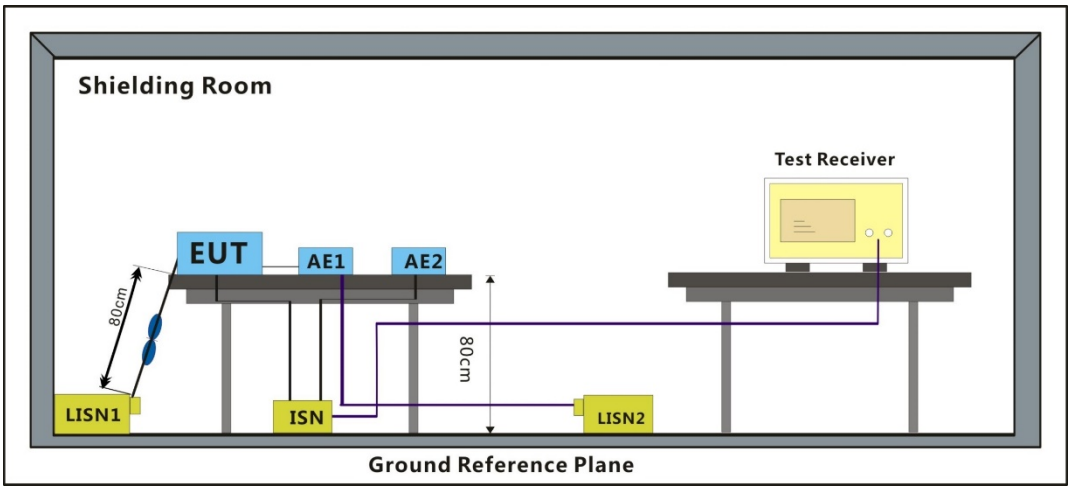
Operating Environment:

Temperature: 26.6 °C Humidity: 55.9 % RH Atmospheric Pressure: 1010 mbar

6.2.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

6.2.3 Test Setup Diagram



6.2.4 Measurement Procedure and Data

Frequency range: 150KHz-30MHz

An initial pre-scan was performed with peak detector.Quasi-Peak or Average measurement were performed at the frequencies with maximized peak emission were detected.

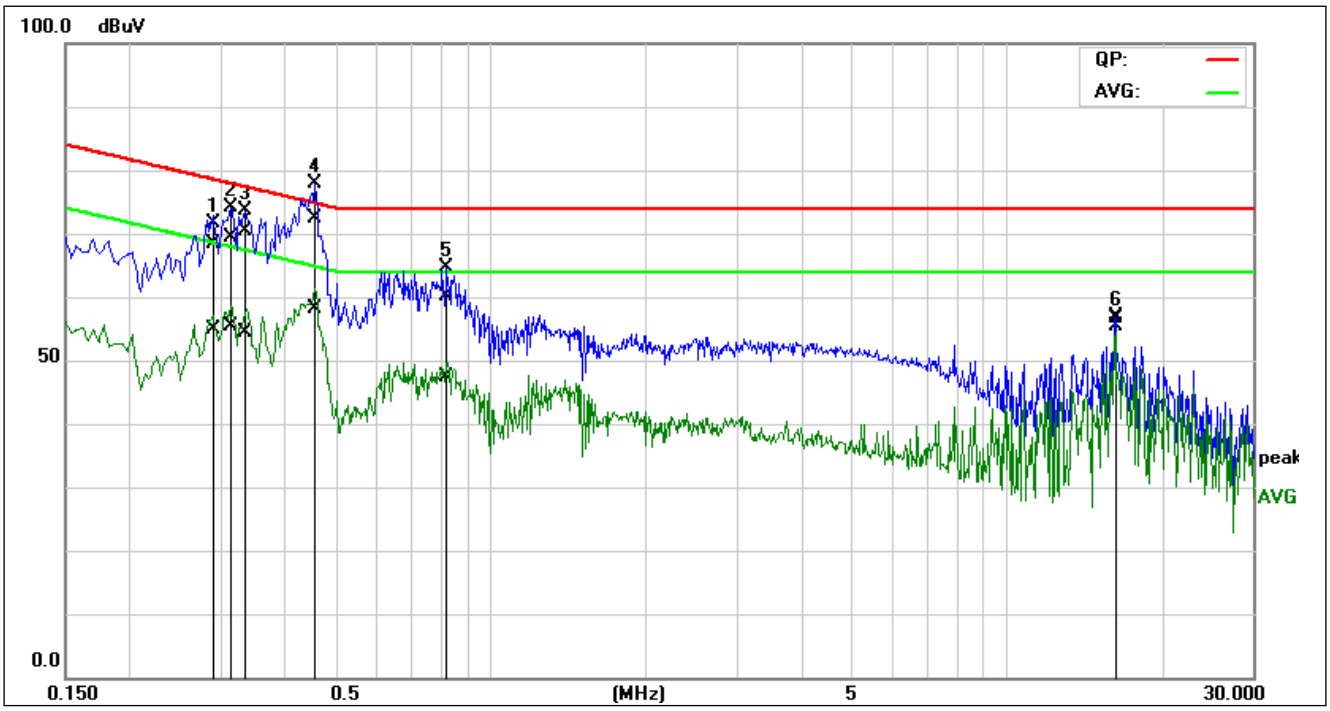
The red line show in graphic is the limit in standard used in this section.

Measured Level = Read level + Cable Loss + LISN Factor

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Test Mode: 00



No.	Frequency	QuasiPeak reading	Average reading	Correction factor	QuasiPeak result	Average result	QuasiPeak limit	Average limit	QuasiPeak margin	Average margin	Remark
	(MHz)	(dBuV)	(dBuV)	(dB)	(dBuV)	(dBuV)	(dBuV)	(dBuV)	(dB)	(dB)	
1	0.2925	48.87	35.19	19.61	68.48	54.80	78.45	68.45	-9.97	-13.65	Pass
2	0.3151	49.83	35.73	19.59	69.42	55.32	77.83	67.83	-8.41	-12.51	Pass
3	0.3318	50.79	34.77	19.60	70.39	54.37	77.41	67.41	-7.02	-13.04	Pass
4*	0.4598	51.93	38.63	19.57	71.50	58.20	73.70	64.70	-3.20	-6.50	Pass
5	0.8212	40.64	27.78	19.51	60.15	47.29	74.00	64.00	-13.85	-16.71	Pass
6	16.2275	36.78	35.37	19.90	56.68	55.27	74.00	64.00	-17.32	-8.73	Pass

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6.3 Radiated Emissions (30MHz-1GHz)

Test Requirement: EN IEC 61000-6-3: 2021

Test Method: CISPR 16-2-3

Limit:

Test Distance:	3m
30MHz-230MHz	40 dB(μV/m) quasi-peak
230MHz-1GHz	47 dB(μV/m) quasi-peak
Detector:	Peak for pre-scan (120kHz resolution bandwidth) 30MHz to 1000MHz

6.3.1 E.U.T. Operation

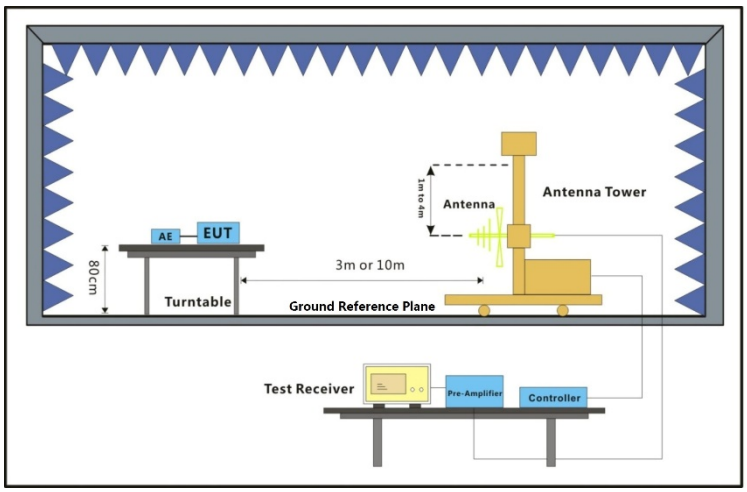
Operating Environment:

Temperature: 23.4 °C Humidity: 42.5 % RH Atmospheric Pressure: 1010 mbar

6.3.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

6.3.3 Test Setup Diagram



6.3.4 Measurement Procedure and Data

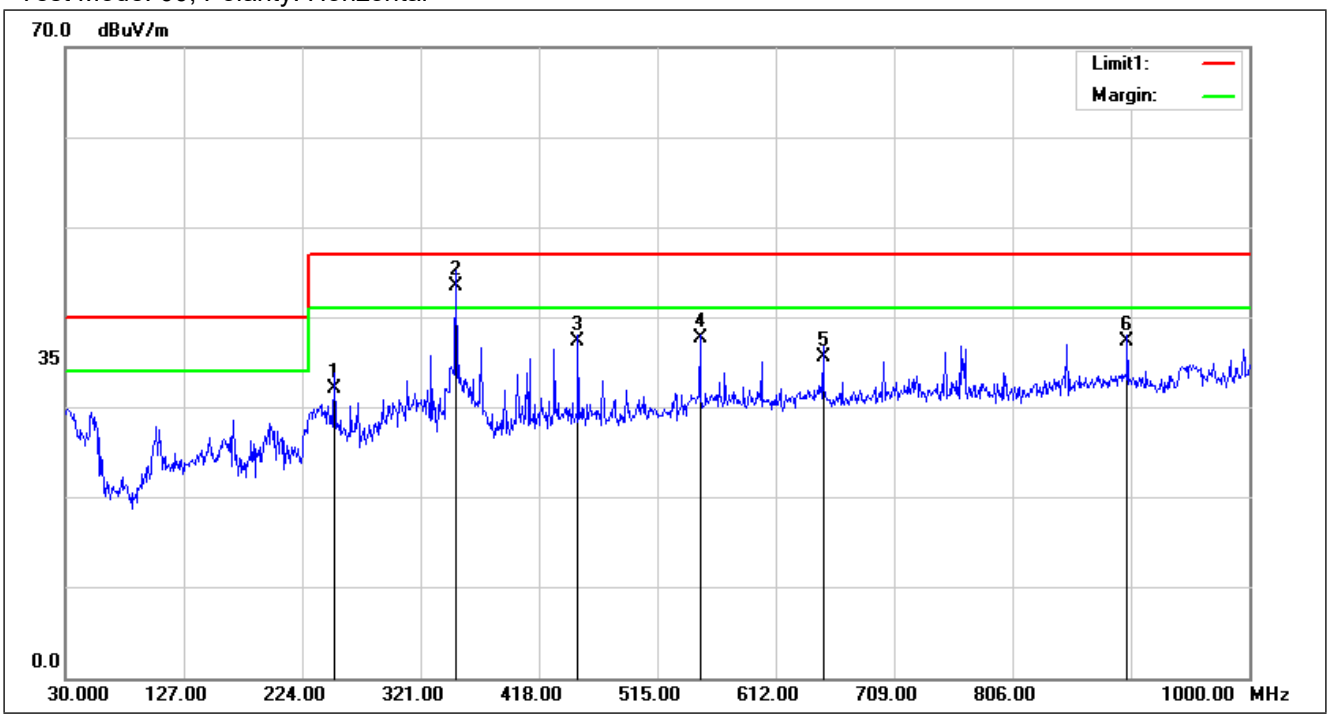
Frequency range: 30MHz-1GHz

An initial pre-scan was performed in the chamber using the spectrum analyser in peak detection mode. Quasi-peak measurements were conducted based on the peak sweep graph. The EUT was measured by BiConiLog antenna with 2 orthogonal polarities.The red line show in graphic is the limit in standard used in this section.

Level=Read Level + Antenna Factor + Cable Loss - Preamp Factor

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Test Mode: 00; Polarity: Horizontal

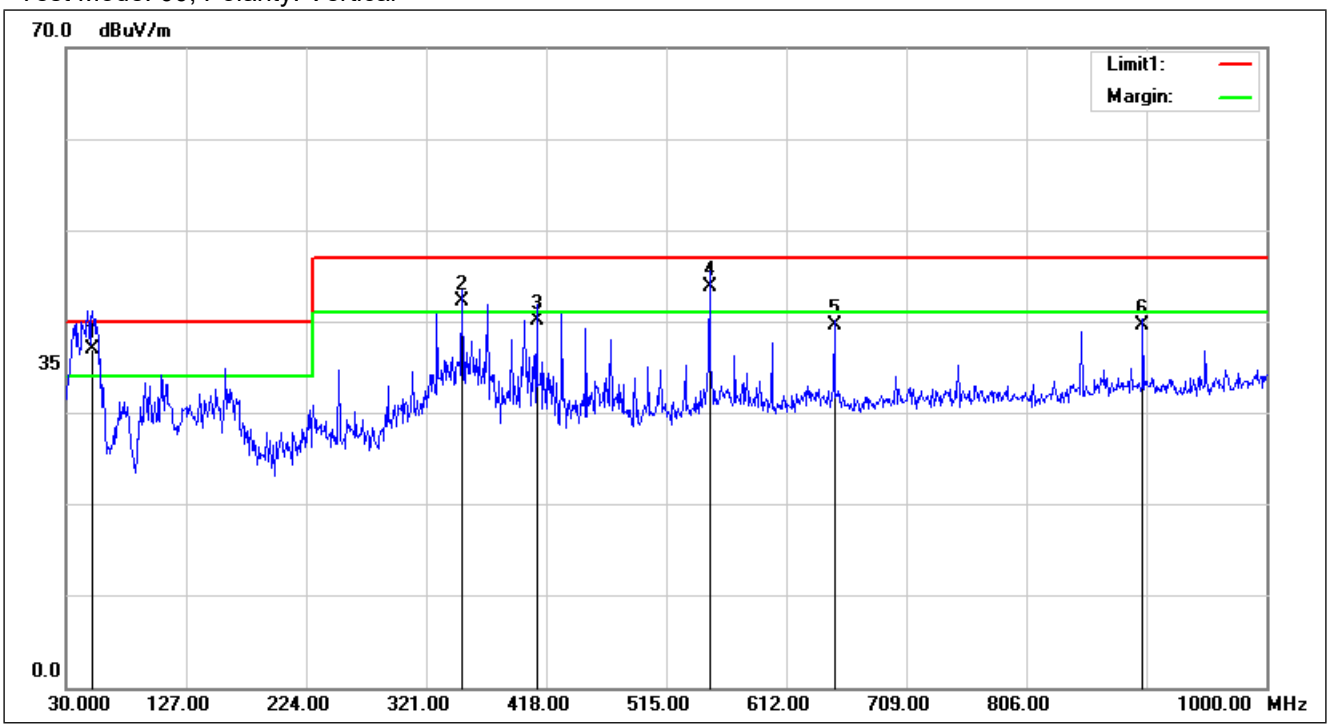


No.	Frequency	Reading	Correct	Result	Limit	Margin	Height	Degree	Remark
	(MHz)	(dBuV)	Factor(dB/m)	(dBuV/m)	(dBuV/m)	(dB)	(cm)	(deg.)	
1	250.1900	12.34	19.78	32.12	47.00	-14.88	100	354	QP
2	350.1000	21.71	21.84	43.55	47.00	-3.45	100	208	QP
3	450.0100	13.11	24.25	37.36	47.00	-9.64	200	13	QP
4	549.9200	10.41	27.22	37.63	47.00	-9.37	100	151	QP
5	650.8000	7.90	27.65	35.55	47.00	-11.45	100	0	QP
6	900.0900	34.88	2.45	37.33	47.00	-9.67	200	0	QP

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Test Mode: 00; Polarity: Vertical



No.	Frequency	Reading	Correct	Result	Limit	Margin	Height	Degree	Remark
	(MHz)	(dBuV)	Factor(dB/m)	(dBuV/m)	(dBuV/m)	(dB)	(cm)	(deg.)	
1	51.3400	19.66	17.30	36.96	40.00	-3.04	99	182	QP
2	350.1000	20.38	21.84	42.22	47.00	-4.78	201	246	QP
3	410.2400	16.43	23.79	40.22	47.00	-6.78	99	362	QP
4	549.9200	16.61	27.22	43.83	47.00	-3.17	201	343	QP
5	650.8000	11.98	27.65	39.63	47.00	-7.37	99	362	QP
6	900.0900	37.18	2.45	39.63	47.00	-7.37	99	362	QP

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6.4 Harmonic Current Emission

Test Requirement: EN IEC 61000-6-3: 2021

Test Method: EN IEC 61000-3-2: 2019+A1:2021

6.4.1 E.U.T. Operation

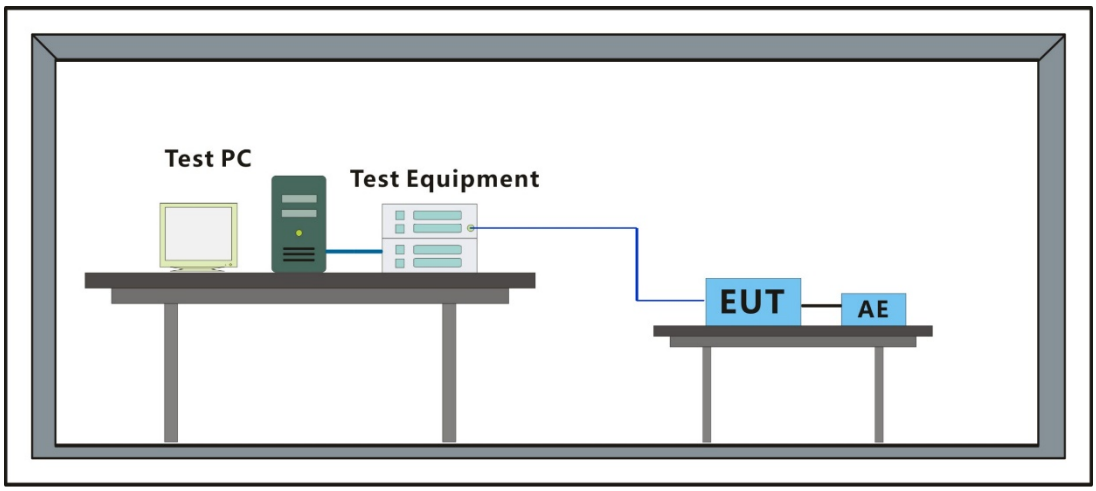
Operating Environment:

Temperature: 24 °C Humidity: 48 % RH Atmospheric Pressure: 1010 mbar

6.4.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

6.4.3 Test Setup Diagram



6.4.4 Measurement Procedure and Data

Frequency Range: 100Hz to 2kHz

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Test Mode: 00

Highest parameter values during test:

V_RMS (Volts):	230.21	Frequency(Hz):	50.00
I_Peak (Amps):	0.144	I_RMS (Amps):	0.021
I_Fund (Amps):	0.009	Crest Factor:	7.034
Power (Watts):	2.0	Power Factor:	0.435

Harm#	Harms(avg)	100%Limit	%of Limit	Harms(max)	150%Limit	%of Limit	Status
2	0.001	1.080	N/A	0.002	1.620	N/A	Pass
3	0.007	2.300	0.3	0.008	3.450	0.2	Pass
4	0.001	0.430	N/A	0.001	0.645	N/A	Pass
5	0.006	1.140	0.5	0.007	1.710	0.4	Pass
6	0.001	0.300	N/A	0.001	0.450	N/A	Pass
7	0.006	0.770	0.8	0.007	1.155	0.6	Pass
8	0.001	0.230	N/A	0.001	0.345	N/A	Pass
9	0.006	0.400	1.5	0.006	0.600	1.0	Pass
10	0.001	0.184	N/A	0.001	0.276	N/A	Pass
11	0.006	0.330	1.7	0.006	0.495	1.2	Pass
12	0.000	0.153	N/A	0.001	0.230	N/A	Pass
13	0.005	0.210	2.5	0.005	0.315	1.7	Pass
14	0.000	0.131	N/A	0.001	0.197	N/A	Pass
15	0.005	0.150	3.4	0.005	0.225	2.3	Pass
16	0.000	0.115	N/A	0.001	0.173	N/A	Pass
17	0.005	0.132	N/A	0.005	0.198	N/A	Pass
18	0.000	0.102	N/A	0.001	0.153	N/A	Pass
19	0.004	0.118	N/A	0.004	0.178	N/A	Pass
20	0.000	0.092	N/A	0.000	0.138	N/A	Pass
21	0.004	0.107	N/A	0.004	0.161	N/A	Pass
22	0.000	0.084	N/A	0.000	0.125	N/A	Pass
23	0.003	0.098	N/A	0.004	0.147	N/A	Pass
24	0.000	0.077	N/A	0.000	0.115	N/A	Pass
25	0.003	0.090	N/A	0.003	0.135	N/A	Pass
26	0.000	0.071	N/A	0.000	0.107	N/A	Pass
27	0.003	0.083	N/A	0.003	0.125	N/A	Pass
28	0.000	0.066	N/A	0.000	0.099	N/A	Pass
29	0.002	0.078	N/A	0.002	0.116	N/A	Pass
30	0.000	0.061	N/A	0.000	0.092	N/A	Pass
31	0.002	0.073	N/A	0.002	0.109	N/A	Pass
32	0.000	0.058	N/A	0.000	0.086	N/A	Pass
33	0.002	0.068	N/A	0.002	0.102	N/A	Pass
34	0.000	0.054	N/A	0.000	0.081	N/A	Pass
35	0.001	0.064	N/A	0.001	0.096	N/A	Pass
36	0.000	0.051	N/A	0.000	0.077	N/A	Pass
37	0.001	0.061	N/A	0.001	0.091	N/A	Pass
38	0.000	0.048	N/A	0.000	0.073	N/A	Pass
39	0.001	0.058	N/A	0.001	0.087	N/A	Pass
40	0.000	0.046	N/A	0.000	0.069	N/A	Pass

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Harm#	Harmonics V-rms	Limit V-rms	% of Limit	Status
2	0.156	0.460	34.00	OK
3	0.447	2.072	21.56	OK
4	0.056	0.460	12.18	OK
5	0.045	0.921	4.87	OK
6	0.041	0.460	9.01	OK
7	0.041	0.690	5.97	OK
8	0.031	0.460	6.74	OK
9	0.030	0.460	6.57	OK
10	0.025	0.460	5.39	OK
11	0.016	0.230	7.02	OK
12	0.018	0.230	7.69	OK
13	0.016	0.230	6.99	OK
14	0.009	0.230	4.12	OK
15	0.010	0.230	4.15	OK
16	0.015	0.230	6.44	OK
17	0.013	0.230	5.71	OK
18	0.016	0.230	6.75	OK
19	0.010	0.230	4.54	OK
20	0.009	0.230	3.85	OK
21	0.012	0.230	5.31	OK
22	0.007	0.230	2.93	OK
23	0.010	0.230	4.36	OK
24	0.007	0.230	2.89	OK
25	0.011	0.230	4.86	OK
26	0.010	0.230	4.55	OK
27	0.008	0.230	3.67	OK
28	0.008	0.230	3.38	OK
29	0.008	0.230	3.66	OK
30	0.007	0.230	2.99	OK
31	0.006	0.230	2.69	OK
32	0.006	0.230	2.42	OK
33	0.007	0.230	2.87	OK
34	0.004	0.230	1.52	OK
35	0.004	0.230	1.89	OK
36	0.003	0.230	1.52	OK
37	0.005	0.230	2.01	OK
38	0.003	0.230	1.13	OK
39	0.004	0.230	1.52	OK
40	0.004	0.230	1.94	OK

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6.5 Voltage Fluctuations and Flicker

Test Requirement: EN IEC 61000-6-3: 2021

Test Method: EN 61000-3-3:2013+A1:2019

6.5.1 E.U.T. Operation

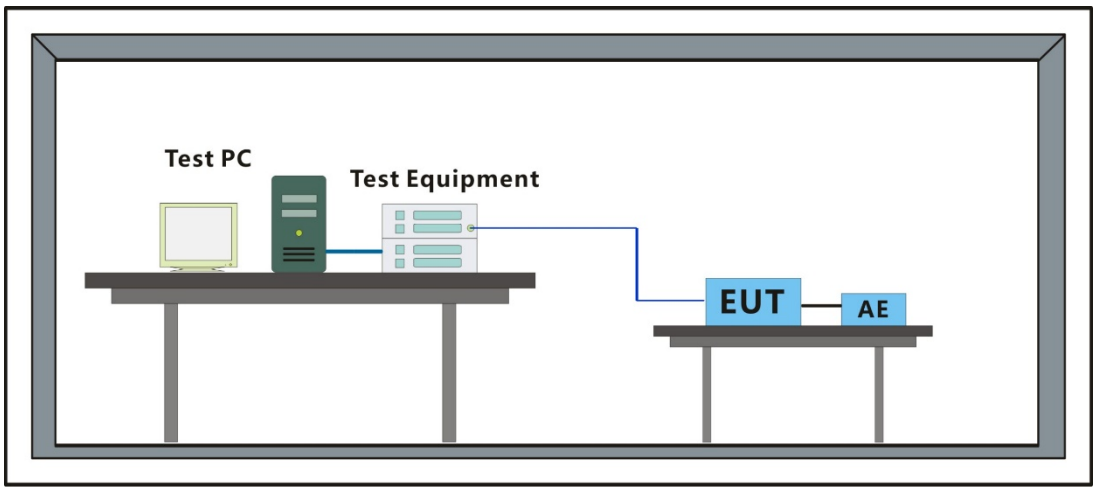
Operating Environment:

Temperature: 24 °C Humidity: 48 % RH Atmospheric Pressure: 1010 mbar

6.5.2 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

6.5.3 Test Setup Diagram



6.5.4 Measurement Procedure and Data

Test Mode: 00

Parameter values recorded during the test:

Vrms at the end of test (Volt):	230.27			
T-max (mS):	0	Test limit (mS):	500.0	Pass
Highest dc (%):	0.00	Test limit (%):	3.30	Pass
Highest dmax (%):	0.00	Test limit (%):	4.00	Pass
Highest Pst (10 min. period):	0.064	Test limit:	1.000	Pass
Highest Plt (2 hr. period):	0.028	Test limit:	0.650	Pass

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7 Immunity Test Results

Performance Criteria Description in EN IEC 61000-6-1:2019

Criterion A	The EUT shall continue to operate as intended during and after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the EUT is used as intended. If the performance level is not specified by the manufacturer, this may be derived from the product description and documentation and what the user may reasonably expect from the equipment if used as intended.
Criterion B	The EUT shall continue to operate as intended after the test. No degradation of performance or loss of function is allowed below a performance level specified by the manufacturer, when the EUT is used as intended. The performance level may be replaced by a permissible loss of performance. However, during the test degradation of performance is allowed but no change of actual operating state or stored data is allowed. If the minimum performance level or the permissible performance loss is not specified by the manufacturer, either of these may be derived from the product description and documentation and what the user may reasonably expect from the equipment if used as intended.
Criterion C	Temporary loss of function is allowed during the test, provided the function is self-recoverable or can be restored by the operation of the controls.

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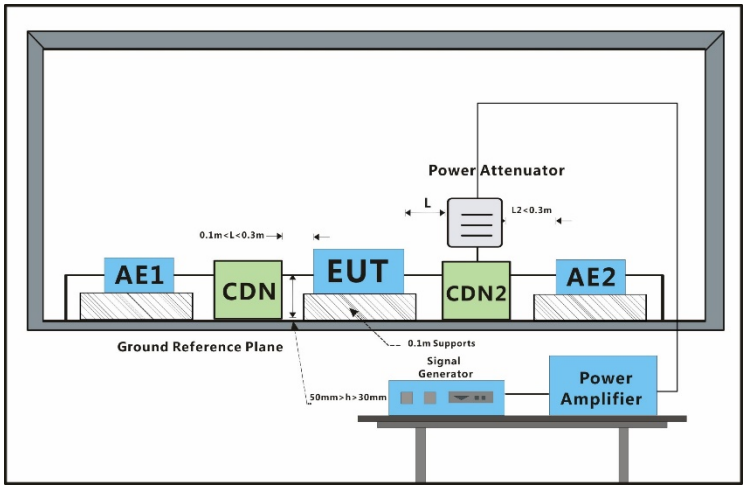
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7.1 Conducted Immunity at AC Mains Power Port (150kHz-80MHz)

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-6:2014

7.1.1 Test Setup Diagram



7.1.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.1.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.1.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

Step Size 1%

Cable port	Level (Vrms)	CDN/Clamp	Dwell time	Result / Observations
AC power port	3(0.15MHz-80MHz)	CDN	3s	A
A: No degradation in the performance of the EUT was observed				

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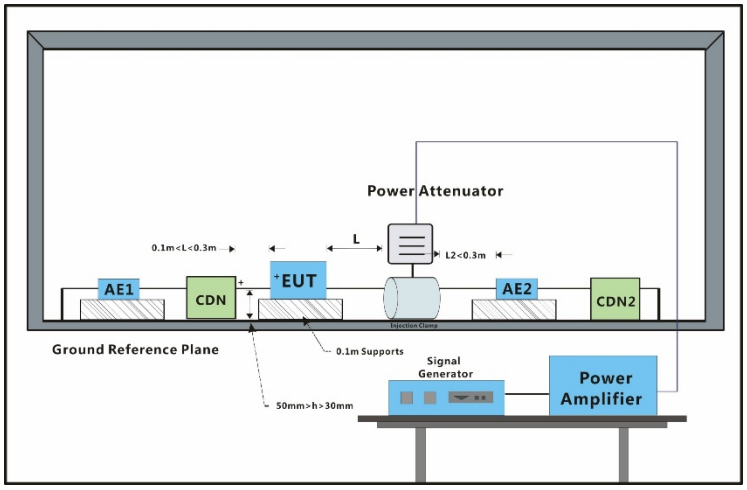
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7.2 Conducted Immunity at Signal Port (150kHz-80MHz)

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-6:2014

7.2.1 Test Setup Diagram



7.2.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.2.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.2.4 Test Condition and Results:

Performance Criterion: A

Frequency Range: 0.15MHz to 80MHz

Modulation: 80%, 1kHz Amplitude Modulation

Step Size 1%

Port	Level (Vrms)	CDN/Clamp	Dwell time	Result / Observations
Signal port	3(0.15MHz-80MHz)	Clamp	3s	A
A: No degradation in the performance of the EUT was observed				

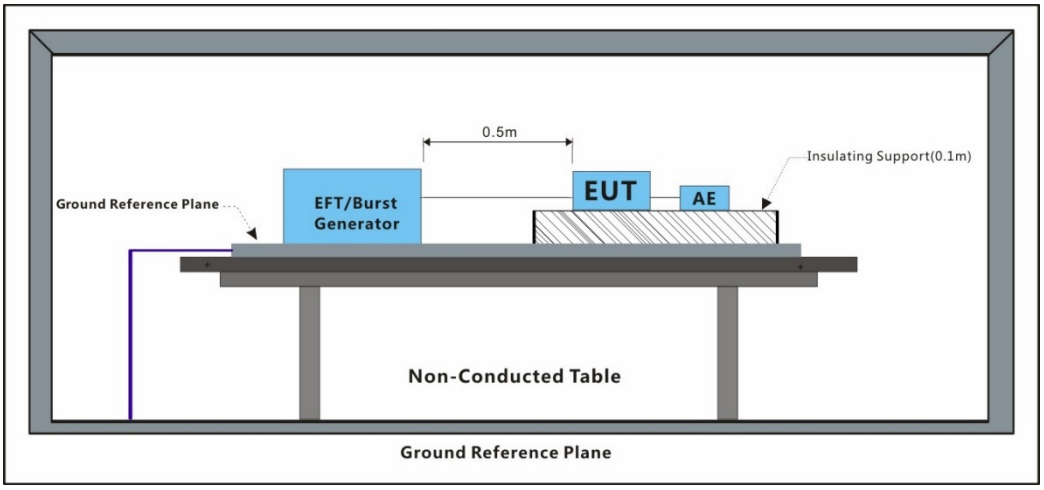
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7.3 Electrical Fast Transients Burst at AC Mains Power Port

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-4:2012

7.3.1 Test Setup Diagram



7.3.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.3.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.3.4 Test Condition and Results:

Performance Criterion: B

Repetition Frequency: 5kHz or 100kHz

Burst Period: 300ms

Test Duration: 2 minute per level & polarity

Test Line	Level (kV)	Polarity	CDN/Clamp	Result / Observations
AC power port	1	+	CDN	A
AC power port	1	-	CDN	A
A: No degradation in the performance of the EUT was observed				

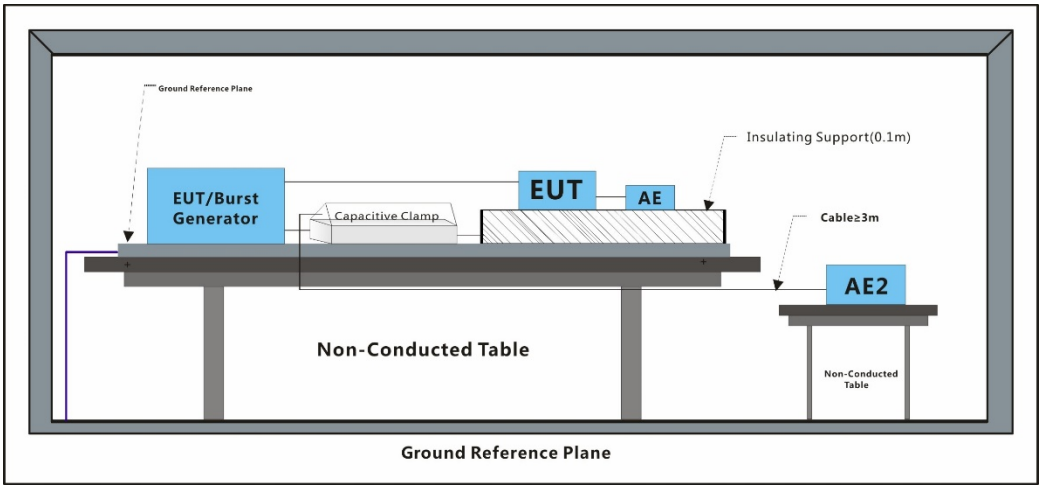
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7.4 Electrical Fast Transients Burst at Signal Port

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-4:2012

7.4.1 Test Setup Diagram



7.4.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.4.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.4.4 Test Condition and Results:

Performance Criterion: B

Repetition Frequency: 5kHz or 100kHz

Burst Period: 300ms

Test Duration: 2 minute per level & polarity

Port	Level (kV)	Polarity	CDN/Clamp	Result / Observations
Signal port	0.5	+	Clamp	A
Signal port	0.5	-	Clamp	A
A: No degradation in the performance of the EUT was observed				

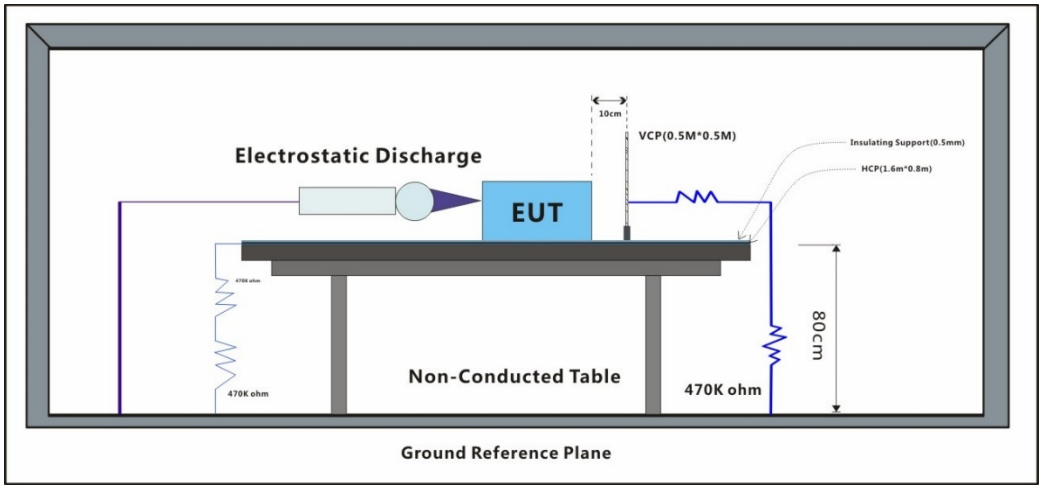
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7.5 Electrostatic Discharge

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-2:2009

7.5.1 Test Setup Diagram



7.5.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 48.0 % RH Atmospheric Pressure: 1010 mbar

7.5.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.5.4 Test Condition and Results:

Performance Criterion: B

Discharge Impedance: 330Ω/150pF

Number of Discharge: Minimum 10 times at each test point

Discharge Mode: Single Discharge

Discharge Period: 1 second minimum

Test Point: 1. All insulated enclosure and seams.

2. All accessible metal parts of the enclosure.

3. All side

Discharge type	Level (kV)	Polarity	Test Point	Result / Observations
Air Discharge	2,4,8	+	1	A
Air Discharge	2,4,8	-	1	A
Contact Discharge	4	+	2	A
Contact Discharge	4	-	2	A
Horizontal	4	+	3	A

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Coupling				
Horizontal Coupling	4	-	3	A
Vertical Coupling	4	+	3	A
Vertical Coupling	4	-	3	A
A: No degradation in the performance of the EUT was observed				

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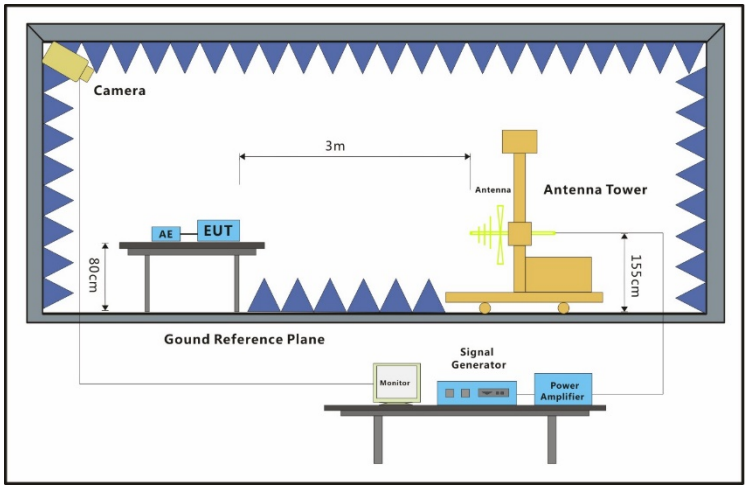
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7.6 Radiated Immunity (80MHz-6GHz)

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN IEC 61000-4-3: 2020

7.6.1 Test Setup Diagram



7.6.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 47.9 % RH Atmospheric Pressure: 1010 mbar

7.6.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.6.4 Test Condition and Results:

Performance Criterion:A

Antenna Polarisation:Vertical and Horizontal

Modulation:1kHz,80% Amp. Mod,1% increment

Frequency Range:80MHz to 1GHz, 1.4GHz to 6GHz

Frequency	Level (V/m)	EUT Face	Dwell time	Result / Observations
80MHz-1GHz	3	Front	3s	A
80MHz-1GHz	3	Back	3s	A
80MHz-1GHz	3	Left	3s	A
80MHz-1GHz	3	Right	3s	A
80MHz-1GHz	3	Top	3s	A
80MHz-1GHz	3	Underside	3s	A
1.4GHz-6GHz	3	Front	3s	A
1.4GHz-6GHz	3	Back	3s	A
1.4GHz-6GHz	3	Left	3s	A
1.4GHz-6GHz	3	Right	3s	A

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1.4GHz-6GHz	3	Top	3s	A
1.4GHz-6GHz	3	Underside	3s	A
A: No degradation in the performance of the EUT was observed				

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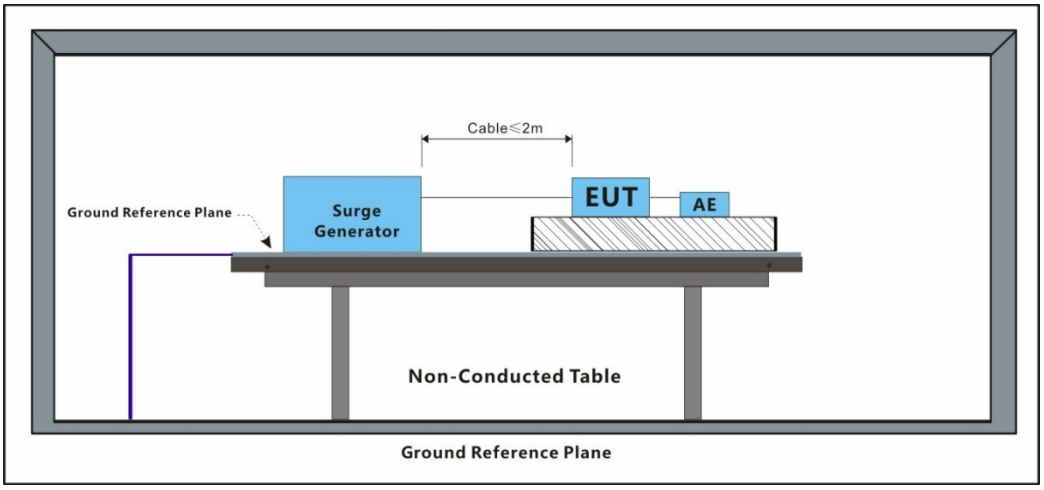
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7.7 Surge at AC Mains Power Port

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-5:2014+A1:2017

7.7.1 Test Setup Diagram



7.7.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.7.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.7.4 Test Condition and Results:

Performance Criterion: B

Interval: 60s between each surge

No. of surges: 5 positive, 5 negative at 0°, 90°, 180°, 270°.

Test Line	Level (kV)	Polarity	Phase (deg)	Result / Observations
L-N	0.5, 1	+	0°	A
L-N	0.5, 1	-	0°	A
L-N	0.5, 1	+	90°	A
L-N	0.5, 1	-	90°	A
L-N	0.5, 1	+	180°	A
L-N	0.5, 1	-	180°	A
L-N	0.5, 1	+	270°	A
L-N	0.5, 1	-	270°	A
A: No degradation in the performance of the EUT was observed				

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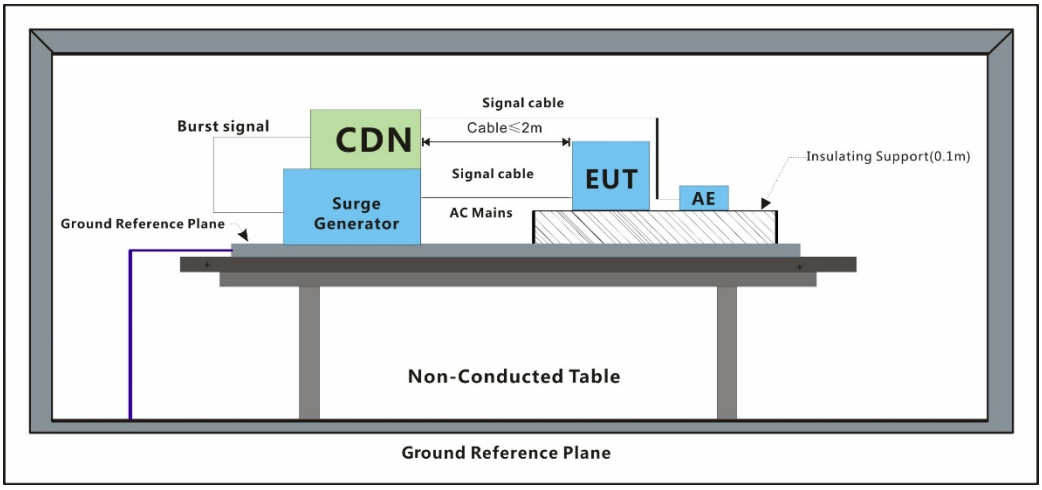
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7.8 Surge at Signal Port

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-5:2014+A1:2017

7.8.1 Test Setup Diagram



7.8.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.8.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.8.4 Test Condition and Results:

Performance Criterion: B

Interval: 60s between each surge

No. of surges: 5 positive, 5 negative

Port	Line	Level (kV)	Polarity	Result / Observations
Signal port	Line-Ground	0.5, 1	+	A
Signal port	Line-Ground	0.5, 1	-	A
A: No degradation in the performance of the EUT was observed				

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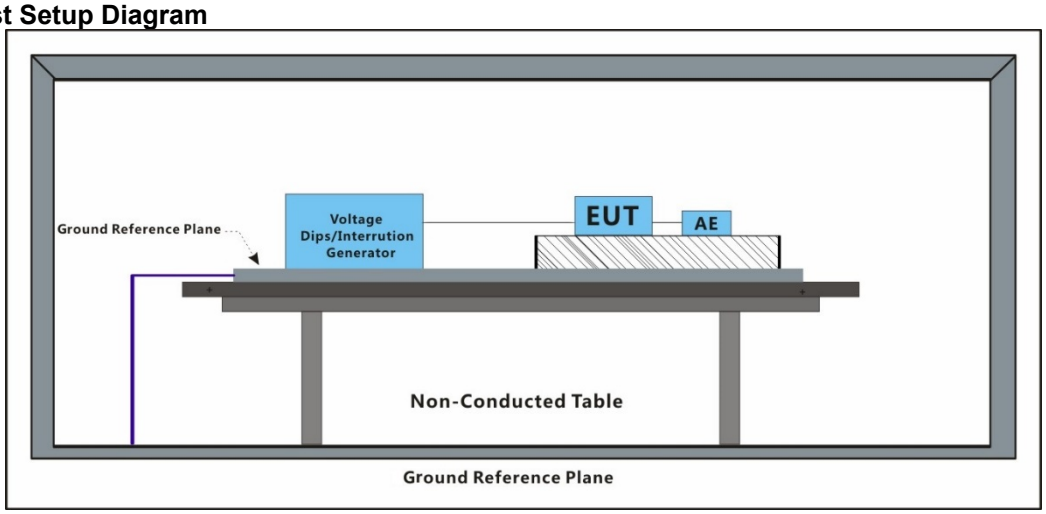
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7.9 Voltage Dips and Interruptions

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN IEC 61000-4-11:2020

7.9.1 Test Setup Diagram



7.9.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.9.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.9.4 Test Condition and Results:

Performance Criterion:

0% of UT (Supply Voltage) for 0.5 Cycle:B;

0% of UT for 1 Cycle:B;

0% of UT for 250 Cycles:C;

70% of UT for 25 Cycles:C;

No. of Dips / Interruptions: 3 per Level

Time between dropout 10s

Level % UT	Phase (deg)	Duration	No. of Dips / Interruptions	Result / Observations
0	0°	0.5 Cycles	3	A
0	180°	0.5 Cycles	3	A
0	0°	1 Cycles	3	A
0	180°	1 Cycles	3	A
70	0°	25 Cycles	3	A
70	180°	25 Cycles	3	A
0	0°	250 Cycles	3	C
0	180°	250 Cycles	3	C

A: No degradation in the performance of the EUT was observed

C: During the test EUT stop working when power supply drop,
After the test by manual operation can work normally.

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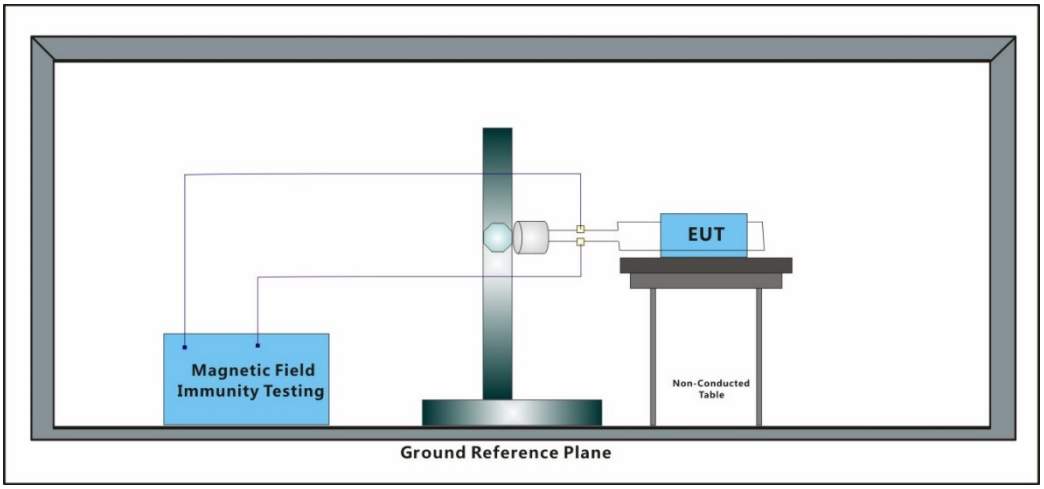
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7.10 Power Frequency Magnetic Field

Test Requirement: EN IEC 61000-6-1: 2019

Test Method: EN 61000-4-8:2010

7.10.1 Test Setup Diagram



7.10.2 E.U.T. Operation

Operating Environment:

Temperature: 23.4 °C Humidity: 44.6 % RH Atmospheric Pressure: 1010 mbar

7.10.3 Test Mode Description

Pre-scan / Final test	Mode Code	Description
Final test	00	Keep EUT working continuously with Auxiliary equipment

7.10.4 Test Condition and Results:

Performance Criterion: A

Frequency	Level (A/m)	Axial	Magnetic Field Type	Result / Observations
50 Hz	3	X	Continue	A
50 Hz	3	Y	Continue	A
50 Hz	3	Z	Continue	A
A: No degradation in the performance of the EUT was observed				

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8 Test Setup Photo

Refer to Appendix - Test Setup Photo for KSEM2204000432AT

9 EUT Constructional Details (EUT Photos)

Refer to Appendix - Photographs of EUT Constructional Details for KSEM2204000432AT

- End of the Report -

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